CHAPTER 8 - Confidence interval estimation

SDS188

2026

1 Properties of estimators

Let:

θ = any parameter

$\hat{\theta}$ = estimator for θ

Examples of parameters: μ, π, σ

Examples of estimators: $\bar{X}$, p, S

1.1 Property 1: Unbiasedness

An estimator is unbiased if its expected value is equal to the parameter.

$$E(\hat{\theta}) = \theta$$

1.2 Property 2: Consistency

An unbiased estimator is consistent if the difference between the estimator and the parameter becomes smaller as the sample size increases.

$$n \uparrow \longrightarrow (\hat{\theta} - \theta) \downarrow$$

In other words, the value of the estimator becomes closer to the true value of the population parameter.

For example,

Consider a population X with $\mu = 75$. As samples are collected from the population of increasing sizes, the value of the sample mean $\bar{X}$ will be closer to 75.

1.3 Property 3: Relative efficiency

For two unbiased estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ for θ, the estimator with the smaller standard error will be a more efficient estimator for θ. A smaller standard error indicates less variation, which means that the estimate will more likely be closer to the parameter.

Formula: (not handled in SDS188)

$$\text{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{Var(\hat{\theta}_2)}{Var(\hat{\theta}_1)}$$

If $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators for θ, the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is larger than 1, only if $Var(\hat{\theta}_2) > Var(\hat{\theta}_1)$. $\hat{\theta}_1$ will thus be a better estimator for θ.

2 Confidence interval for the population mean where the population standard deviation is known

$(1 - \alpha)$% CI for μ:

$$\bar{x} \pm z_{\frac{\alpha}{2}} \times \frac{\sigma}{\sqrt{n}}$$

Values needed:

point estimate	$\bar{x}$
confidence level	α
standard deviation	σ
sample size	n

Calculations:

$z_{\frac{\alpha}{2}}$	=NORM.S.INV($1 - \frac{\alpha}{2}$) OR =ABS(NORM.S.INV($\frac{\alpha}{2}$))	$1-\frac{\alpha}{2} \longrightarrow$  OR $\frac{\alpha}{2} \longrightarrow$ 
standard error ($\frac{\sigma}{\sqrt{n}}$)	=σ/SQRT(n)	σ  n

3 Confidence interval for the population mean where the population standard deviation is unknown

$(1-\alpha)$% CI for μ:

$$\bar{x} \pm t_{\frac{\alpha}{2},(n-1)} \times \frac{s}{\sqrt{n}}$$

Values needed:

point estimate (can be calculated)	$\bar{x}$
confidence level	α
standard deviation (can be calculated)	s
sample size (can be calculated)	n

Calculations:

$\bar{x}$	=AVERAGE(data)	x_1 Σ+ … x_n Σ+ ↘ 7 $\bar{x},\bar{y}$
s	=STDEV.S(data)	x_1 Σ+ … x_n Σ+ ↘ 8 Sx,Sy
df	=n-1	n TAN √x̄ 1 RAND = DISP
$t_{\frac{\alpha}{2},(n-1)}$	=T.INV($1-\frac{\alpha}{2}$,df) OF =ABS(T.INV($\frac{\alpha}{2}$,df))	$df \longrightarrow$ ↗ INV M+) df,t⇄P 2 LN $1-\frac{\alpha}{2}$ RAND = DISP OR $df \longrightarrow$ ↗ INV M+) df,t⇄P 2 LN $\frac{\alpha}{2}$ RAND = DISP +/− E
standard error ($\frac{s}{\sqrt{n}}$)	=s/SQRT(n)	s SIN ÷ 1/x n ↘ TAN √x̄

4 Confidence interval for population proportion

$(1-\alpha)$% CI for π:

$$p \pm z_{\frac{\alpha}{2}} \times \sqrt{\frac{p(1-p)}{n}}$$

Values needed:

point estimate	$p = \frac{X}{n}$
confidence level	α
sample size	n

Calculations:

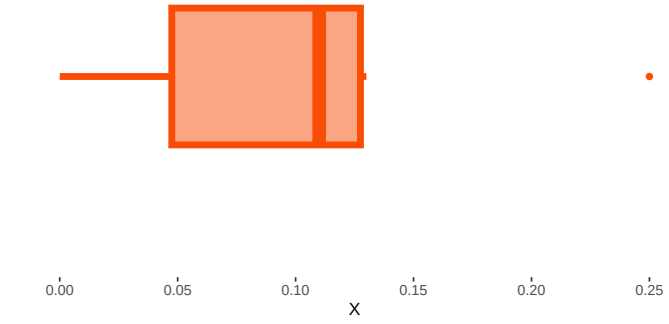
X	=COUNTIF(data,"whatever is defined as a success")
n	=COUNTA(data)
p	=X/n
$z_{\frac{\alpha}{2}}$	=NORM.S.INV($1-\frac{\alpha}{2}$) OR =ABS(NORM.S.INV($\frac{\alpha}{2}$))
standard error ($\sqrt{\frac{p(1-p)}{n}}$)	=SQRT((p*(1-p))/n)

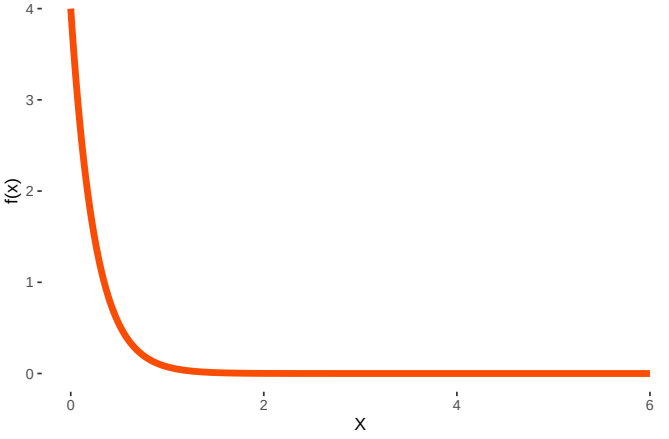
5 Bootstrap estimation

Let $X \sim$ exponential$(\lambda = 4)$

x	0.16	0.03	0.50	0.23	0.46	0.02	0.07	0.89

Distribution of X:





Since X is exponentially distributed, the data has a positively skewed distribution. Therefore, the above mentioned formulas cannot be used. (The assumptions are not met)

Select 1000 samples with replacement from the original sample **x**:

x	0.16	0.03	0.50	0.23	0.46	0.02	0.07	0.89

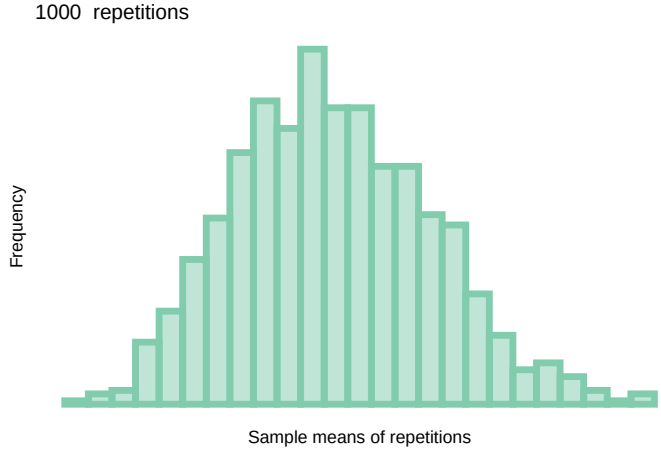
1	0.13	0.42	0.10	0.49	0.12	0.12	0.25	0.12
2	0.42	0.12	0.42	0.25	0.12	0.25	0.03	0.00
3	0.25	0.12	0.25	0.00	0.25	0.03	0.03	0.42
4	0.49	0.49	0.13	0.10	0.42	0.13	0.42	0.25
5	0.10	0.42	0.13	0.12	0.25	0.13	0.03	0.49
6	0.12	0.49	0.42	0.12	0.12	0.00	0.10	0.10
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
995	0.13	0.42	0.10	0.13	0.25	0.42	0.03	0.12
996	0.25	0.42	0.42	0.00	0.10	0.49	0.42	0.49
997	0.10	0.00	0.00	0.42	0.10	0.42	0.49	0.12
998	0.12	0.49	0.13	0.42	0.03	0.00	0.00	0.49
999	0.03	0.12	0.49	0.10	0.13	0.12	0.12	0.25
1000	0.42	0.25	0.49	0.49	0.12	0.49	0.25	0.03

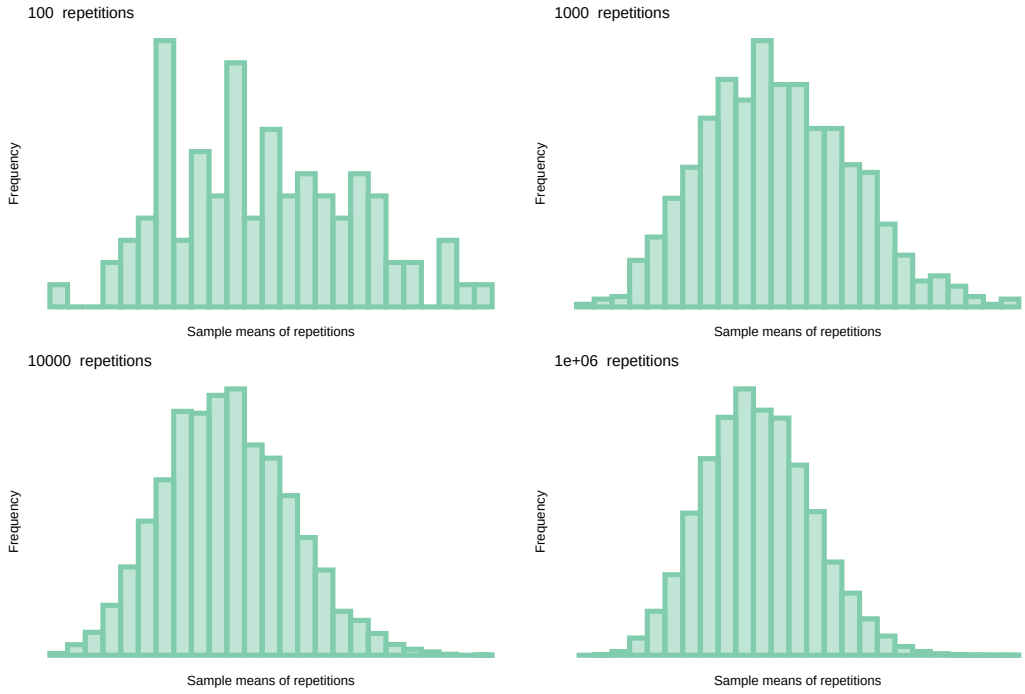
The sample mean ($\bar{x}$) is calculated for each of the 1000 samples

The last column contains the **sample means** for each sample.

	1	2	3	4	5	6	7	8	$\bar{x}$
1	0.13	0.42	0.10	0.49	0.12	0.12	0.25	0.12	0.22
2	0.42	0.12	0.42	0.25	0.12	0.25	0.03	0.00	0.20
3	0.25	0.12	0.25	0.00	0.25	0.03	0.03	0.42	0.17
4	0.49	0.49	0.13	0.10	0.42	0.13	0.42	0.25	0.30
5	0.10	0.42	0.13	0.12	0.25	0.13	0.03	0.49	0.21
6	0.12	0.49	0.42	0.12	0.12	0.00	0.10	0.10	0.18
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
995	0.13	0.42	0.10	0.13	0.25	0.42	0.03	0.12	0.20
996	0.25	0.42	0.42	0.00	0.10	0.49	0.42	0.49	0.32
997	0.10	0.00	0.00	0.42	0.10	0.42	0.49	0.12	0.21
998	0.12	0.49	0.13	0.42	0.03	0.00	0.00	0.49	0.21
999	0.03	0.12	0.49	0.10	0.13	0.12	0.12	0.25	0.17
1000	0.42	0.25	0.49	0.49	0.12	0.49	0.25	0.03	0.32

The graph below represents the histogram of all 1000 sample means. Note that as m increases, the distribution of the sample means appear to be approximately normally distributed. But as m increases by too much, the distribution will not change much. THUS a value of 1000 is enough.





5.1 Bootstrap confidence interval

To determine the limits of the bootstrap confidence interval, the **sample means** should be ordered from smallest to largest. Determine the position of the lower limit with $m \times \frac{\alpha}{2}$ and the upper limit with $m \times (1 - \frac{\alpha}{2})$.

For the data above:

Let $\alpha = 0.05$ Lower limit position:

$$1000 \times 0.025 = 25$$

The 25th value is 0.08. Upper limit position:

$$1000 \times 0.975 = 975$$

The 975th value is 0.31.

We therefore have 95% confidence that the true population mean is between 0.08 and 0.31.

In this case we have access to the population and since $X \sim$ exponential$(\lambda = 4)$, the true population mean is $\frac{1}{4} = 0.25$. This value is between 0.08 and 0.31.

5.2 Extra: R-code

For interest sake, here is the R-code (This software only gets used in Statistics 214)

```
X<-rexp(n = 8,rate = 4)
B<-1000
boot.mat<-matrix(0,nrow = B,ncol = 8)
boot.xbar<-rep(0,B)
for(i in 1:B){
  ind.vec.0<-1:length(X)
  for(j in 1:length(X)){
    ind.samp<-sample(x = ind.vec.0,size = length(X),replace = T)
    boot.mat[i,j]<-X[ind.samp[j]]
  }
  boot.xbar[i]<-mean(boot.mat[i,])
}
sort.xbar<-sort(boot.xbar)
```

6 Questions

6.1 Question 1

A paper manufacturer has a production process that is continuously operated during a production shift. The paper is expected to have an average length of 11 inches and the standard deviation of the

length is 0.04 inches. Samples are collected from time to time to determine if the average length is equal to 11 inches or to determine if something went wrong during the process.

A random sample of 100 pages are collected, of which the average length is 10.999 inches. A 95% confidence interval for the population mean length is $10.99116 \leq \mu \leq 11.00684$.

Is it true that you do not know for sure that the population mean is between 10.99116 and 11.00684 inches. Explain.

6.2 Question 2

Assume that the population this sample comes from is normally distributed.

1	2	3	4	5	29

1. Construct a 99% confidence interval for the population mean.
2. Replace the 29 with 6 and construct another 99% confidence interval for the population mean.
3. What effect does an extreme value have on the general confidence interval? Choose the correct answer:

	The extreme value in the original data causes a decrease in the value of the sample mean and a decrease in the sample standard deviation which will result in a narrower confidence interval.
	The extreme value in the original data causes a decrease in the value of the sample mean and an increase in the sample standard deviation which will result in a wider confidence interval.
	The extreme value in the original data causes an increase in the value of the sample mean and a decrease in the sample standard deviation which will result in a narrower confidence interval.
	The extreme value in the original data causes an increase in the value of the sample mean and an increase in the sample standard deviation which will result in a wider confidence interval.

6.3 Question 3

If $n = 200$ and $X = 70$, construct a 95% confidence interval for the population proportion.

6.4 Question 4

Consider a population of 1000, and assume that the standard deviation is 20. What sample size is required if the sample is collected without replacement if you have a confidence level of 95% and a sampling error of ± 5.

6.5 Question 5

If $\bar{x} = 79$, $s = 29$ and $n = 81$, assume that the population is normally distributed and construct a 90% confidence intercal for the population mean μ.

6.6 Question 6

Consider constructing a 99% confidence interval for the mean, and the assumption of a normally distributed population is not met. A sample of 8 are selected from the population without replacement, from which 1000 samples of 8 are collected repeatedly with replacement. What are the next steps in constructing a 99% confidence interval for μ?

6.7 Question 7

Determine the required sample size with 90% confidence level for the estimation of the population proportion, π, within an interval of 0.006. Previous information indicates that $\pi = 0.2$.

6.8 Question 8

If $\bar{x} = 75$, $s = 24$, $n = 36$ and $N = 200$, construct a 95% confidence interval for μ if the sample is selected without replacement.

6.9 Question 9

What sample size is needed to estimate the population mean within an error of ± 3 and $\sigma = 20$ if a confidence level of 95% is preferred?